

Fisher Forecasts for Euclid, LiteBIRD, and CMB-S4

David A. Besemer^{1,*}

¹*Independent Researcher*

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We present Fisher matrix forecasts for next-generation cosmological observatories (Euclid, LiteBIRD, CMB-S4) under the SDGFT framework. We show that the tensor-to-scalar ratio prediction $r = 0.013$ and the neutrino isocurvature fraction $\beta_{iso} = 1/36$ will be tested at the 5 – sigma level within the next decade, providing a test of the theory.

I. INTRODUCTION

II. THEORETICAL FORMULATION

Here we formulate the core mathematical relations governing the system:

$$\sigma(r) \approx 0.001 \quad (\text{LiteBIRD sensitivity}), \quad \sigma(\beta_{iso}) \approx 0.005 \quad (\text{CMB-S4}) \quad (1)$$

III. FISHER MATRIX FORMULATION

We construct the Fisher information matrix for the cosmological parameter set and show that the forecasts constrain the Horndeski parameters with high precision.

IV. CONCLUSION

We project the sensitivity of upcoming cosmological surveys to the unique predictions of SDGFT, focusing on the CMB polarization sector.

Next-generation experiments will be able to distinguish SDGFT from Starobinsky inflation and LCDM, making the theory fully testable.

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- [3] D. A. Besemer, `sdgft` Python package, <https://github.com/david-besemer/sdgft> (2026).

* david.besemer@sdgft.org